Percent and Ratio Reasoning

CIVIL-SERVICE · numerical-ability · BoardPrep lesson v2

Official examination scope

CSC Examination Announcement No. 03 s. 2026 lists basic operations, number sequence, and word problems under Numerical Ability. This lesson practices percent and ratio applications within basic operations and word problems. It does not estimate item frequency or section weight; check current CSC notices before an exam.

Learning objectives

- Identify the whole, part, rate, and units before choosing an operation in a word problem.
- Solve percent-of, percent-change, ratio, proportion, and unit-rate questions with visible
- steps.
- Check the base, ratio order, units, and estimated size of an answer before selecting it.

1. Percent of a base

Prerequisite: convert a percent p% to the decimal p/100. Label the whole as the base, the percentage as the rate, and the result as the part. Then use part = rate × base. If the part and rate are known but the base is not, solve base = part ÷ rate. For a discount, calculate the discount from the original price and subtract it; do not add the discount rate to the sale price to reverse the calculation. Keep currency or item units beside the numbers.

2. Percent change

For an increase or decrease, calculate the change first and compare it with the original amount: percent change = change ÷ original × 100%. The original amount is the base because the question asks how much the value changed relative to where it began. State whether the result is an increase or decrease. If a question gives the final amount after a percentage change, set up an equation using the final rate of the original rather than applying the percentage to the final amount.

3. Ratios and rates

Write a ratio in the order named in the question, then scale both terms by the same factor. A unit rate divides one quantity by its reference units, such as cases per hour. For combined work, add rates only when workers act independently at steady rates on the same kind of task; multiply the combined rate by time to get output. Convert units to one common basis before comparing.

Essential relationships

Before choosing an answer, estimate its size and check the units. A percentage greater than 100% can be valid, but a discount amount should not exceed its original base. For a ratio, verify that both parts were scaled equally and that the order matches the question. If the result is a rate, write the numerator and denominator units explicitly. These checks catch a misplaced decimal, reversed ratio, or wrong percent base.

- part = rate as a decimal x base
- percent change = (new - original) / original x 100%
- equivalent ratio: multiply or divide every term by the same nonzero factor
- unit rate = quantity / reference units

Find a percent of an amount

Question: What is 18% of 250?
1. Write 18% as 0.18.
2. Multiply 0.18 x 250 = 45.
Answer: 18% of 250 is 45.

Check: Ten percent of 250 is 25 and eight percent is 20; 25 + 20 = 45.

Find a percent increase

Question: An amount rises from 500 to 575. What is the percent increase?
1. Change = 575 - 500 = 75.
2. Divide by the original amount: 75 / 500 = 0.15.
3. Convert to a percent: 0.15 x 100% = 15%.
Answer: The increase is 15% of the original amount.
Check: 500 x 1.15 = 575. Using 575 as the denominator would answer a different question.

Undo a discount

Question: A price after a 15% discount is 680. What was the price before the discount?
1. The sale price is 100% - 15% = 85% of the original.
2. Write 85% as 0.85 and solve original x 0.85 = 680.
3. Original = 680 / 0.85 = 800.
Answer: The original price was 800.
Check: A 15% discount on 800 is 120; 800 - 120 = 680. Adding 15% of 680 would use the wrong base.

Combine two steady work rates

Question: Worker A processes 36 files per hour and worker B processes 24 files per hour. If they work independently at steady rates for 2.5 hours, how many files do they process together?
1. Add the compatible rates: 36 + 24 = 60 files per hour.
2. Multiply by time: 60 files per hour × 2.5 hours = 150 files.
3. Check that hours cancel and the remaining unit is files.
Answer: Together they process 150 files under the stated steady, independent-work assumptions.
Check: If the workers share a bottleneck or work sequentially, adding their rates would not be justified.

Common misconceptions

- Using 18 instead of 0.18 when multiplying by a percent.
- Dividing percent change by the new amount instead of the original base.
- Adding a discount percent to the discounted price to reverse the discount.
- Scaling only one term of a ratio or comparing rates with different time units.

Practice patterns

- For percent problems, label the original base before calculating the part or change.
- For ratio problems, preserve the order and scale every term by the same factor.
- For rate and work problems, write units, state the constant-rate assumption, then estimate the
- result.

Official scope references

Official scope: CSC Examination Announcement No. 03 s. 2026: 09 August 2026 CSE-PPT Scope — Philippine Civil Service Commission — https://csc.gov.ph/phocadownload/userupload/erpo/announcements/2026/ExamAnn_2026_03_Conduct%20of%2009%20Aug%202026%20CSE-PPT_orig%20signed.pdf

Instructional references

Instructional reference: Prealgebra 2e, Section 6.2: Solve General Applications of Percent — OpenStax, Rice University — https://openstax.org/books/prealgebra-2e/pages/6-2-solve-general-applications-of-percent

Instructional reference: Prealgebra 2e, Section 6.5: Solve Proportions and Their Applications
— OpenStax —
https://openstax.org/books/prealgebra-2e/pages/6-5-solve-proportions-and-their-applications

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